Activity 5:
Generating a sequence of all *rational numbers* greater than 1

Pat says: There must be more numbers between 1 and infinity than between 0 and 1. *Is Pat right? Explain.*

Task A: Think about *rational numbers* greater than 1

One way to generate the *rational numbers* greater than 1 is to redo Activity 4 and make a few small changes. But there is a clever way that is much less work.	*Rational numbers* are all numbers that can be made by dividing two *integers*.

In Activity 4 we made fractions a/b where a is less than b. What kinds of numbers do we get if we had b/a instead?

Explain.

Task B: Programming all rational numbers greater than 1

Train a robot named `Divides 1` *that takes a number and gives a bird 1 divided by that number. The pictorial instructions are at the end if you need help.*

Use the robots from Activity 4 to make all the proper fractions. Run `Divides 1` *with the nest with all the fractions.*

What is the result?

Connect `Match Maker` with the result. Think about what the robots are producing.

Sally says that something is fishy here. She says: “There must be the same number of rational numbers from 0 to 1 as there are from 1 to 2. And from 2 to 3. And 3 to 4. So there are an infinite number of *intervals* of the same size as 0 to 1. So there must be an infinite times more rational numbers greater than 1 than between 0 and 1.” Do you agree or disagree with Sally? Why?

Activity 5, Task B: Training `Divides 1`

Task B: Programming all rational numbers greater than 1